Start

Initiate Federated Learning for Smartwatch Health Data
(Privacy-Preserving)

IoT Smartwatch 1

IoT Smartwatch 2

IoT Smartwatch 3

IoT Smart Health Watch
(Heart Rate ,BP,Temp,SpO2,Activity)

Data Collection Layer

Identify Sensitive Data
(Health & Personal Parameters)

Differential Privacy Layer
(Add Controlled Noise using DP)

Privacy-Protected Data

Local Model Training
(On Device)

Send Noisy Update
(Weights, Gradients)

Global Federated Model
(Aggregated from All Devices)

Privacy-Safe Output

Health Alert&Recommendation

Personalized Local Model
(Each Device Fine-Tunes the Global Model Locally)

Global Model Update / Local Fine-Tuning